

# Bayesian Hill Mixture Models for Marketing Mix Modeling: Synthetic Benchmarks, Component Resolvability, and Real-Data Evaluation

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## Abstract

Marketing Mix Models (MMMs) typically assume that all consumers respond to advertising with a single Hill saturation curve, but real audiences are heterogeneous. We study a Bayesian mixture of  $K$  Hill curves implemented in NumPyro, with an ordering reparameterisation, post-hoc relabeling, and label-invariant convergence diagnostics designed to make mixture posteriors interpretable.

We report three experiments. (i) A controlled synthetic benchmark shows that mixture models improve expected log pointwise predictive density (ELPD-LOO) whenever the true data-generating process is heterogeneous, and incur only a small penalty otherwise. (ii) A *component resolvability* study varies the cosine distance between true Hill curves and identifies an empirical threshold of  $\bar{d}_{\cos} \approx 0.10$ , below which posteriors collapse to fewer effective components. (iii) A real-data benchmark on three Conjura organisations across three verticals (beauty & fitness, home & garden, toys & hobbies) shows that Mixture  $K=3$  improves ELPD-LOO over single-Hill in every organisation, by 222 to 308 units, with the gain primarily in predictive density rather than point error.

The benchmark convergence threshold ( $\hat{R} \leq 1.05$ ) that 44 of 45 synthetic fits satisfy fails uniformly for Mixture  $K=3$  on real data, where every fit reports  $\max \hat{R} \geq 1.05$  and several show severe chain mixing failures ( $\max \hat{R}$  up to 3.6, tail ESS per chain as low as 1–6). Re-using the resolvability axis on the real posteriors shows that this gap is organisation-dependent: Home & Garden sits at the resolvability boundary, Toys & Hobbies is consistent with three resolvable components but with chain-mixing caveats, and Beauty & Fitness shows strong multi-modality in the posterior coupled with influential observations. We report these three regimes directly rather than collapsing them into a single summary. Code, configurations, and per-fit artifacts are released for reproduction.

**Keywords:** Marketing Mix Modeling, Bayesian inference, Hill function, mixture models, label switching, component resolvability

## 1 Introduction

Marketing Mix Modeling (MMM) enables organisations to quantify the effectiveness of advertising spend and to optimise budget allocation. The canonical formulation fits a single response curve per channel on aggregate time-series data, combining geometric (or Weibull) adstock for carryover with a Hill function for saturation [Jin et al., 2017, Chan and Perry, 2017]. We refer to this baseline as *Standard MMM*.

Standard MMM assumes that all consumers respond to a given channel with the same saturation curve. In practice, audiences are heterogeneous: heavy versus light buyers, loyalists versus

switchers, and geographically distinct segments exhibit qualitatively different response patterns. A single aggregate curve is a weighted average of these latent responses, and the aggregate estimate can be misleading even when in-sample fit looks good. This paper studies a Bayesian mixture of Hill curves as a principled extension of Standard MMM that makes the latent heterogeneity explicit, and asks two empirical questions: how much heterogeneity must be present before the mixture can recover it, and how well do the recovery guarantees that hold on synthetic data transfer to field data?

### Contributions.

1. A controlled synthetic benchmark comparing single-Hill and Bayesian mixture-of-Hill models under known ground truth, with predictive, latent-recovery, and convergence metrics.
2. A *component resolvability* study that systematically varies the cosine distance between true Hill curves and traces the recovered Shannon effective component count as a function of true separation, identifying an empirical threshold below which the posterior collapses to fewer components.
3. A real-data benchmark on three Conjura organisations across three verticals, paired with a re-use of the resolvability axis on the real posteriors. The combination shows that mixture-of-Hill improves predictive density uniformly across organisations but that publication-ready convergence is organisation-dependent in a way the resolvability framework can diagnose.

**Roadmap.** Section 2 situates the contribution against the MMM, Bayesian segmentation, and mixture-identification literatures. Section 3 specifies the model and the label-invariant diagnostic workflow. Section 4 consolidates the inference configuration, evaluation metrics, and publication-ready convergence criterion used throughout the paper. Sections 5–7 report the three benchmarks. Section 8 discusses what worked and what did not, Section 9 states limitations, and Section 10 describes the released artifacts.

## 2 Related Work

**Bayesian MMM with adstock and saturation.** The combination of geometric adstock with Hill saturation under a Bayesian likelihood is the canonical modern MMM formulation [Jin et al., 2017, Chan and Perry, 2017]. Subsequent open-source implementations (PYMC-MARKETING, MERIDIAN) have made this family practical at industrial scale; we keep our model in the same family so that the mixture extension is a minimal, drop-in change rather than a separate architecture. We deliberately do not extend the dynamics: Dew et al. [2024] argue that nonlinear and time-varying effects in MMM are jointly not identified from typical observational time series, so apparent saturation can be an artefact of misspecified dynamics. Their critique motivates our *controlled* treatment of nonlinearity, in which heterogeneity rather than dynamics is the source of curve shape.

**Heterogeneous response and segmentation.** The marketing literature has long emphasised consumer heterogeneity [Wedel and Kamakura, 2000, Allenby and Rossi, 1998]. Hierarchical Bayes and latent-class models are the standard tools at the individual or panel level. MMM, by contrast, operates on aggregated time series and typically suppresses heterogeneity. Mixture-of-Hill is one of the smallest extensions that brings explicit latent segments back into the aggregate model without leaving the standard MMM family.

**Mixture identifiability and label switching.** Finite-mixture posteriors are non-identified under permutation of component labels [Stephens, 2000, Frühwirth-Schnatter, 2006]. The label-switching problem makes per-component summaries meaningless without an additional convention, and breaks standard split- $\hat{R}$  diagnostics [Vehtari et al., 2021]. The two standard remedies—ordering constraints and post-hoc relabeling [Papastamoulis, 2014]—trade off detailed balance against interpretability. We combine both with an additional label-invariant diagnostic (Section 3.1) and treat them as complementary rather than substitutable.

### 3 Model Specification

Let  $x_t$  denote aggregated marketing spend at time  $t$  and  $y_t$  the observed outcome (purchases, revenue, or a related target). Figure 1 shows the model in plate notation. The model is implemented in NumPyro [Phan et al., 2019] with JAX acceleration.

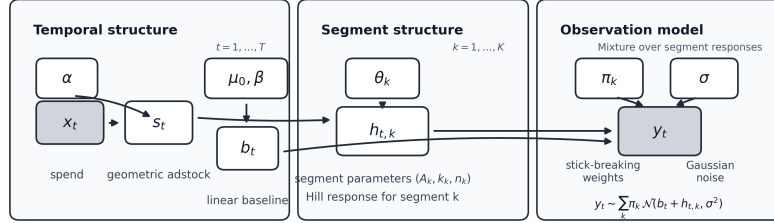


Figure 1: Graphical model. Observed variables are shaded; plates indicate replication over time steps  $t$  and mixture components  $k$ .

**Adstock transformation.** Geometric decay captures carryover from previous periods:

$$s_t = x_t + \alpha \cdot s_{t-1}, \quad s_0 = 0, \quad \alpha \sim \text{Beta}(2, 2). \quad (1)$$

**Hill saturation.** Each mixture component  $k \in \{1, \dots, K\}$  has its own Hill response:

$$f_k(s) = A_k \cdot \frac{s^{n_k}}{\lambda_k^{n_k} + s^{n_k}}, \quad (2)$$

with maximum effect  $A_k$ , half-saturation point  $\lambda_k$ , and slope  $n_k \sim \text{LogNormal}(\log 1.5, 0.4)$ .

**Mixture likelihood.** The observation model is a Gaussian mixture over components:

$$y_t \sim \sum_{k=1}^K \pi_k \cdot \mathcal{N}(\mu_0 + \beta t + f_k(s_t), \sigma^2), \quad (3)$$

where  $\pi \sim \text{Dirichlet}(\mathbf{1}_K)$  and  $\mu_0, \beta$  capture intercept and trend.

**Automatic prior scaling.** To accommodate diverse data scales, priors on  $A_k$ ,  $\mu_0$ , and  $\sigma$  are computed from training-data statistics:  $A_k \sim \text{LogNormal}(\log(0.3 \cdot \text{range}(y)), 0.8)$ ,  $\mu_0 \sim \mathcal{N}(\bar{y}, 2\sigma_y)$ ,  $\sigma \sim \text{HalfNormal}(\sigma_y)$ .

### 3.1 Identifiability and Label Switching

A finite mixture is invariant under permutations of component labels: any reordering of  $(\pi_k, A_k, \lambda_k, n_k)$  yields the same likelihood. As a result, posterior summaries computed per component are meaningless without an additional convention, and standard split- $\hat{R}$  on raw component parameters cannot distinguish genuine non-mixing from label permutations across chains [Stephens, 2000, Frühwirth-Schnatter, 2006].

**Within-MCMC ordering.** We impose  $\lambda_1 < \lambda_2 < \dots < \lambda_K$  via cumulative-sum reparameterization with positive increments  $\delta_j \sim \text{LogNormal}(\log(s_{\max}/(K+1)), 0.7)$ . This restricts the posterior to one canonical labelling but can distort posterior geometry near the boundary.

**Post-hoc relabeling.** We additionally implement unconstrained sampling followed by post-hoc reordering: each posterior draw is permuted so that  $\lambda_1 < \lambda_2 < \dots < \lambda_K$  [Stephens, 2000, Papastamoulis, 2014]. This preserves detailed balance and exposes interpretable per-component summaries.

**Label-invariant diagnostics.** We compute  $\hat{R}$  and effective sample sizes on three groups of quantities [Vehtari et al., 2021]:

1. *Joint log-likelihood*  $\sum_t \log \sum_k \pi_k \phi(y_t \mid \mu_t^{(k)}, \sigma)$ , which is label-invariant by construction.
2. *Scalar parameters*  $(\alpha, \sigma, \mu_0, \beta)$  that are not affected by label permutations.
3. *Relabeled component parameters*, after the post-hoc reordering above.

A model is treated as effectively converged when all three pass; this is the criterion we use to report a *publication-ready* fit. Concretely, a fit is publication-ready when (i) the No-U-Turn sampler produces zero divergences, zero maximum-tree-depth saturations, and a minimum chain-level BFMI of at least 0.30; and (ii) every retained mixing diagnostic satisfies  $\hat{R} \leq 1.01$  with both bulk and tail effective sample size of at least 100 per chain. The mixing diagnostics retained depend on the model: for single-Hill we use raw parameter  $\hat{R}/\text{ESS}$ ; for mixtures we replace the raw component-parameter diagnostics with the label-invariant diagnostics above and additionally require the relabeled component parameters to satisfy the same thresholds. These thresholds match the conservative defaults of Vehtari et al. [2021] and ensure that “pass” is a strict criterion rather than a soft one.

## 4 Experimental Setup

**Inference.** All fits use the No-U-Turn Sampler (NUTS) with two chains, init strategy `numpyro.infer.init_to_mcmc` and diagonal mass adaptation. Warmup and sampling lengths and the target acceptance probability are model-dependent; on synthetic data we use (600, 600) for single-Hill on the  $K = 1$  and  $K = 2$  DGPs and (800, 800) on the  $K = 3$  DGP, (900, 900) for Mixture  $K = 2$ , and (1,600, 1,200) for Mixture  $K = 3$ . On real data the same per-model schedule is used with (600, 600), (900, 900), and (1,600, 1,200) respectively. Target acceptance is increased for larger mixtures to control divergences. The complete configuration is reproduced in Appendix A.

**Data splits.** Each time series is divided into a 75/25 train/test split. Holdout predictions inherit the adstock state accumulated during training and are evaluated at their absolute time index rather than reset to zero, so that adstock decay and the trend term are propagated forward consistently.

**Evaluation metrics.** We report (i) expected log pointwise predictive density estimated by Pareto-smoothed importance-sampling leave-one-out cross-validation (ELPD-LOO), with the count of  $\hat{k}_i > 0.7$  observations reported alongside as a reliability flag; (ii) test-set mean absolute percentage error (MAPE), continuous ranked probability score (CRPS), and 90% predictive coverage; and (iii) for synthetic conditions, latent-mean normalised RMSE against the known DGP latent.

**Publication-ready convergence.** The criterion defined in Section 3.1 is applied identically across all experiments: zero divergences, zero max-tree-depth saturations,  $\text{BFMI} \geq 0.30$  on every chain,  $\hat{R} \leq 1.01$ , and bulk/tail ESS per chain  $\geq 100$  on the retained mixing diagnostics (raw parameters for single-Hill; label-invariant joint log-likelihood plus scalar non-component parameters, together with relabeled component parameters, for mixtures).

**Component summaries.** For mixture fits we additionally report posterior mixture weights  $\hat{\pi}_k$ , the Shannon effective component count  ${}^1D = \exp(-\sum_k \hat{\pi}_k \log \hat{\pi}_k)$  [Hill, 1973], and the mean pairwise cosine distance between active Hill response curves. The latter is computed on a fit-adaptive saturation grid that covers the active components’ half-saturation range, making it independent of how the prior scale was set.

## 5 Synthetic Benchmark

### 5.1 Data-Generating Processes

We evaluate the models across three data-generating processes (DGPs): (1) *Single* ( $K = 1$ ), a single Hill response; (2) *Mixture*  $K = 2$ , a two-component mixture; and (3) *Mixture*  $K = 3$ , a three-component mixture. Each DGP uses  $T = 200$  observations and we run five random seeds per condition, so the design contains  $3 \times 3 \times 5 = 45$  fits.

### 5.2 Results

**Convergence.** Under the benchmark threshold  $\hat{R} \leq 1.05$  on raw parameters for single-Hill and on label-invariant diagnostics for mixtures, 44 of 45 synthetic fits cleared the threshold (Figure 2). The single exception was the Mixture  $K = 3$  model on the  $K = 2$  DGP at seed 1, which reported a max  $\hat{R} \approx 1.5$  and is flagged as a benchmark failure rather than silently included. Standard single-Hill diagnostics agreed with the mixture-specific label-invariant diagnostics on every other synthetic fit. The stricter publication-ready criterion ( $\hat{R} \leq 1.01$ , ESS  $\geq 100$  per chain, plus the HMC clauses of Section 3.1) is more demanding: 33 of 45 fits cleared it, with the remaining cases recorded as warnings (typically  $\hat{R}$  in the range  $[1.01, 1.05]$ ). We retain the strict criterion as the headline figure for real data in Section 7, but the synthetic Section 5 plots use the looser benchmark threshold so that the comparison against published MMM benchmarks is on equal footing.

**Predictive performance.** Figures 3 and 4 report ELPD-LOO and holdout CRPS. When the true DGP is homogeneous ( $K = 1$ ), the three models tie within standard error. When heterogeneity is present, mixtures gain substantially in ELPD ( $\Delta \approx +50$  on  $K = 2$ ,  $\Delta \approx +45$  on  $K = 3$ ). CRPS

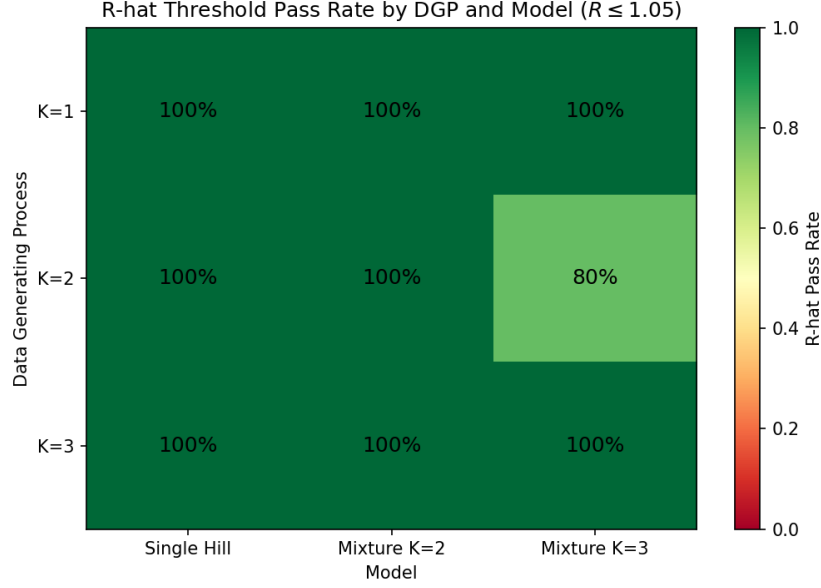


Figure 2: Benchmark convergence pass rate ( $\hat{R} \leq 1.05$  on the retained mixing diagnostics) by DGP and model on the synthetic benchmark. Eight cells reach 100%; the Mixture  $K = 3$  model on the  $K = 2$  DGP reaches 80% (4 of 5 seeds), with seed 1 failing at max  $\hat{R} \approx 1.5$ .

gains are noticeably smaller than ELPD gains ( $\Delta \approx 0.1$  on  $K = 2$ ,  $\Delta \approx 0.4$  on  $K = 3$ ), indicating that the bulk of the improvement is in probabilistic density rather than in point prediction.

**Latent recovery.** Figure 5 shows latent-mean nRMSE on the test split. The pattern mirrors ELPD: mixtures roughly halve nRMSE under  $K = 2$  ( $0.14 \rightarrow 0.07$ ) and reduce it by a factor of  $\approx 2.4$  under  $K = 3$  when the model is matched to the true  $K$  ( $0.085 \rightarrow 0.035$ ). This shows that the ELPD gains correspond to genuinely better recovery of the latent response, not just better marginal calibration.

## 6 Component Resolvability

The previous section shows that mixtures help when heterogeneity exists, but does not tell us *how much* heterogeneity is required for the posterior to actually separate components. We address this with a controlled study.

### 6.1 Design

We construct nine DGP profiles by varying the spacing  $\delta$  between true Hill half-points around a fixed centre  $c$ , holding amplitude and slope fixed. For  $K_{\text{true}} = 1$  we use one anchor profile; for  $K_{\text{true}} \in \{2, 3\}$  we use four profiles each, spanning low to extreme separation. Each profile is run with five seeds and fit with both Mixture  $K = 2$  and Mixture  $K = 3$ , for  $9 \times 2 \times 5 = 90$  fits in total. The separation axis  $\bar{d}_{\text{cos}}$  and recovery axis  ${}^1D$  are the cosine-separation and Shannon-effective-count measures defined in Section 4; here  $\bar{d}_{\text{cos}}$  is computed on the *true* component curves and  ${}^1D$  on the posterior mixture weights.

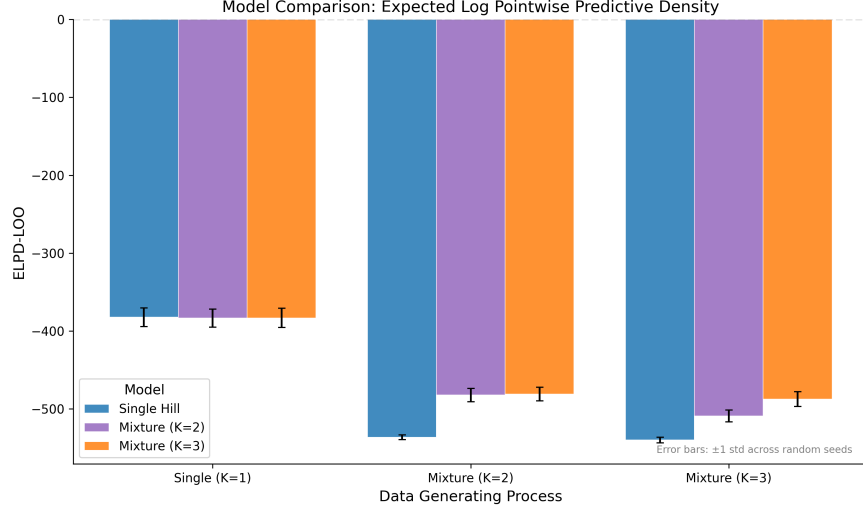


Figure 3: ELPD-LOO by DGP and model on the synthetic benchmark. Higher is better. Mixtures improve substantially when  $K \geq 2$ .

## 6.2 Results

Figure 6 plots  $^1D$  against  $\bar{d}_{\cos}$  across all 90 fits. Three regimes are visible:

- For  $\bar{d}_{\cos} \lesssim 0.05$ , both models collapse toward the trivial value: even when  $K_{\text{true}} \geq 2$ , the posterior cannot resolve more than  $\approx 1.5$ – $1.7$  effective components.
- Between  $\bar{d}_{\cos} \approx 0.05$  and  $0.3$ , the recovered count rises sharply with separation: this is the resolvable transition region.
- For  $\bar{d}_{\cos} \gtrsim 0.3$ , the recovered count saturates near  $K_{\text{true}}$ . Mixture  $K = 2$  saturates near 2 by construction; Mixture  $K = 3$  tracks the true  $K$  and approaches 3 on  $K = 3$  DGPs.

**Failure modes.** The publication-ready convergence criterion fails most frequently in two regimes. First, under-specified fits at low separation: Mixture  $K = 2$  on  $K_{\text{true}} = 3$  DGPs at low  $\delta$  shows publication-fail rates of 0.6–0.8. Second, at the highest separation under  $K_{\text{true}} = 3$ , Mixture  $K = 3$  exhibits a non-trivial failure rate (0.6 at the most extreme profile), suggesting that the strict diagnostic criterion is sensitive to posterior geometry even when the components are far apart.

**Practical implication.** Combined with the ELPD results of Section 3, the resolvability picture suggests that mixture-of-Hill models can be used reliably when the cosine separation between true components exceeds  $\approx 0.1$ . Below this threshold, the posterior is consistent with a smaller effective  $K$ , and the modeller should expect both lower convergence pass rates and very limited information gain over a single-Hill baseline.

## 7 Real-Data Benchmark

### 7.1 Dataset and Organisation Selection

We use the public Conjura MMM dataset, which contains daily spend across nine advertising channels and several purchase-related target variables for a panel of e-commerce organisations.

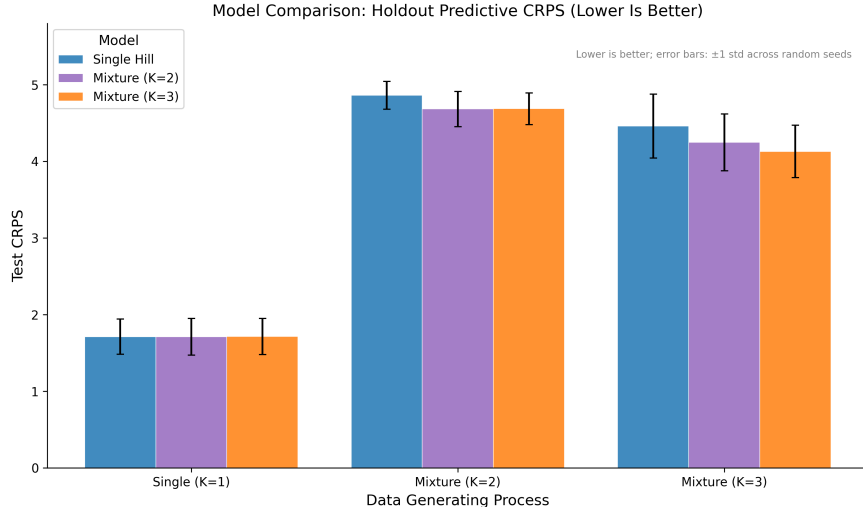


Figure 4: Holdout CRPS by DGP and model on the synthetic benchmark. Lower is better. Gains are smaller in magnitude than ELPD gains.

Following the per-organisation modelling convention of MMM practice, we treat each organisation as an independent time series and do not share parameters across organisations.

To select organisations for the benchmark, we screen all candidates with at least 200 daily observations and at least two active spend channels (non-zero on more than 10% of days), and rank them by a composite score combining series length, spend coefficient of variation, spend–target correlation, and channel breadth. We pick three top-ranked organisations from three different verticals to support a modest cross-vertical claim. The selected organisations are summarised in Table 1.

Table 1: Selected Conjura organisations for the real-data benchmark.

| Vertical         | $T$ (days) | Active channels | Spend–target $r$ | Total spend (USD M) |
|------------------|------------|-----------------|------------------|---------------------|
| Toys & Hobbies   | 1,375      | 5               | 0.95             | 3.3                 |
| Beauty & Fitness | 924        | 8               | 0.96             | 5.2                 |
| Home & Garden    | 1,642      | 5               | 0.80             | 2.1                 |

## 7.2 Predictive Performance

Each organisation is fit with single-Hill, Mixture  $K = 2$ , and Mixture  $K = 3$  under three seeds, giving 27 fits in total; inference and evaluation follow Section 4.

Table 2 reports ELPD-LOO, holdout MAPE, holdout CRPS, and holdout 90% coverage averaged across seeds. The key observation is that *Mixture  $K = 3$  achieves the best ELPD-LOO in every organisation*, with improvements of +222 (beauty & fitness), +308 (home & garden), and +275 (toys & hobbies) over the single-Hill baseline. Mixture  $K = 2$  is intermediate.

A caveat on the magnitude of these gaps. The standard errors in Table 2 are per-fit PSIS-LOO standard errors of the individual ELPD point estimates, not the proper paired difference `se_diff` between two models on the same observations [Vehtari et al., 2017]. Computing `se_diff` requires pointwise log-likelihood matrices that we did not retain. As a coarse upper bound on a paired SE,



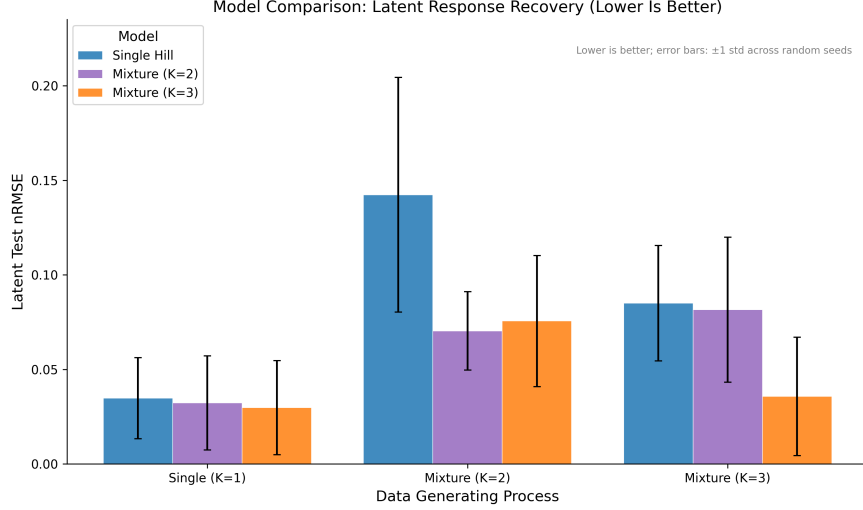


Figure 5: Latent-mean nRMSE by DGP and model. Lower is better. Mixtures recover the latent response more accurately when heterogeneity is present.

treating the two models as independent gives  $\sqrt{86^2 + 48^2} \approx 99$  (beauty & fitness),  $\sqrt{63^2 + 48^2} \approx 79$  (home & garden), and  $\sqrt{89^2 + 91^2} \approx 127$  (toys & hobbies), under which the  $K = 3$  vs. single-Hill gaps are roughly 2–4 such SE in magnitude—meaningful but not as dramatic as the per-fit SE alone might suggest. The seed-level SDs in Figure 7 are Monte Carlo variability across NUTS rerun, not data-level uncertainty, and should be interpreted as a stability check rather than as a substitute for the missing paired SE.

Figure 7 visualises the same comparison. The Mixture  $K = 3$  advantage is consistent in sign across organisations, while the absolute scale of ELPD-LOO varies by an order of magnitude with series length and target variance.

The relationship between ELPD and the other metrics is heterogeneous. On *home & garden* the rank ordering of models is consistent across ELPD, CRPS, and MAPE: the mixture improves all three. On *toys & hobbies*, ELPD and CRPS both improve, while MAPE is non-monotonic. On *beauty & fitness*, ELPD improves substantially but MAPE and CRPS *deteriorate* slightly for  $K = 3$ , and the model remains visibly underconfident: 90% coverage rises from 0.33 (single) to 0.54 ( $K = 3$ ) but stays well below the nominal level. The ELPD gain on this organisation therefore reflects a localised improvement in predictive density rather than a uniform predictive win; this should not be read as “ $K = 3$  predicts better on beauty & fitness”. We return to this in the discussion.

To make concrete what the mixture is doing, Figure 8 shows the posterior response curves for the home & garden organisation under all three models on a single representative seed. Single-Hill recovers one slowly-saturating curve. Mixture  $K = 2$  resolves two components with weights 0.93 and 0.07, in which the small minority component carries a much higher saturation ceiling. Mixture  $K = 3$  uses two active components with weights 0.84 and 0.14 and a third component that collapses to weight 0.02 (labelled *inactive* in the legend). The qualitative two-mode structure is robust to whether  $K$  is set to 2 or 3, which is consistent with the posterior effective component count reported below.

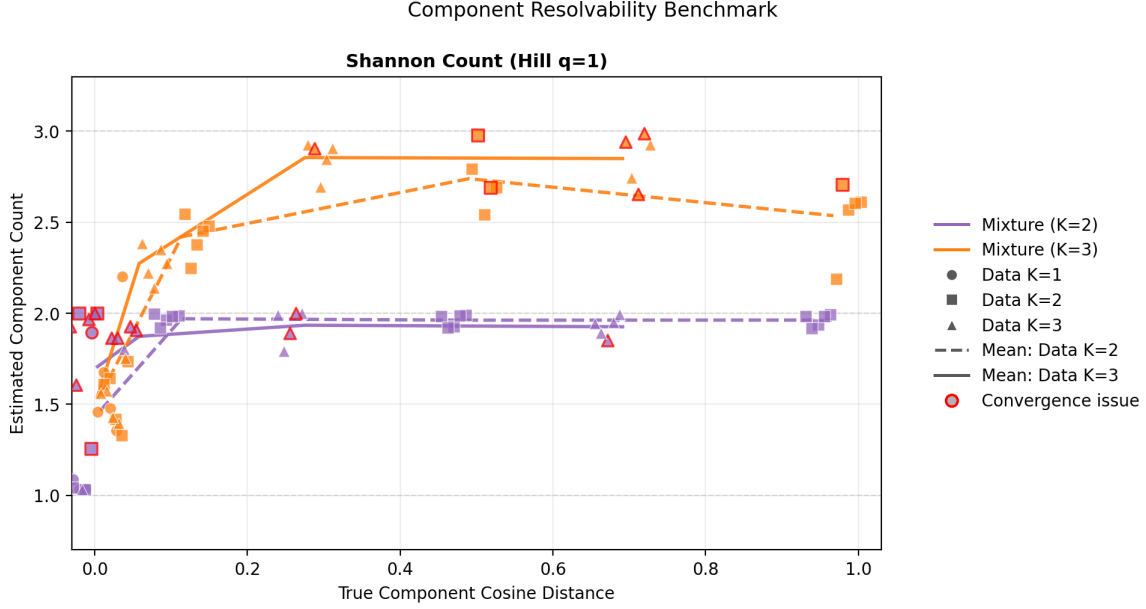


Figure 6: Component resolvability: Shannon effective count of the posterior versus mean pairwise cosine distance of the true Hill curves. Red-outlined points indicate cases that failed the publication-ready convergence criterion.

### 7.3 Convergence on Real Data

Table 3 summarises the rate at which each (organisation, model) cell passes the publication-ready convergence criterion (HMC diagnostics plus label-invariant and relabeled  $\hat{R}$ ) across the three seeds. Two patterns are stark.

First, *Mixture  $K = 3$  passes the publication-ready criterion in 0/9 fits* on real data, despite the matched- $K$  synthetic fits passing in 5/5 runs at matched  $K = 3$  (Section 5). The Mixture  $K = 3$  failures on real data are also not all the same kind: Home & Garden chains hit the strict threshold only marginally, whereas Beauty & Fitness and Toys & Hobbies report max  $\hat{R}$  of 2.4–3.6 and 1.8, with tail ESS per chain of 1–6 and 23–34 respectively, and large numbers of max-tree-depth saturations. We therefore treat the three organisations’  $K = 3$  fits as qualitatively different failure modes rather than a single “0/9” bucket (Section 7.4). Second, the Shannon effective count  $^1D_{\text{post}}$  is at most 2.72 (Toys & Hobbies  $K = 3$ ), while in two of the three organisations it stays below 1.8, suggesting that the draws rarely use the full  $K = 3$  capacity in those cases.

The high Pareto- $k$  count for the beauty & fitness Mixture  $K = 2$  fit (15 observations on average) is consistent with a moderately misspecified model on data with influential outliers, and points to robustifying the likelihood as a useful next step.

### 7.4 Where Real Organisations Sit on the Resolvability Axis

The synthetic resolvability study (Section 6) characterises recovery as a function of *true* cosine separation between component curves. Real data has no ground truth, but we can compute the same quantity from each fit’s *posterior* component summary: the mean pairwise cosine distance between the active Hill curves implied by the posterior means  $(A_k, \lambda_k, n_k)$ , on a fit-adaptive saturation grid. Plotting the recovered Shannon effective count against this posterior separation places the real organisations onto the same plane as the synthetic resolvability fits (Figure 9).

Table 2: Real-data predictive performance, averaged across three seeds. ELPD-LOO is reported with the per-fit PSIS-LOO standard error in parentheses (averaged across seeds); higher is better. MAPE/CRPS lower is better; 90% coverage closer to 0.9 is better. Bold marks the best ELPD-LOO per organisation.

| Organisation     | Model           | ELPD-LOO ( $\pm$ SE)              | Test MAPE | Test CRPS | 90% cov. |
|------------------|-----------------|-----------------------------------|-----------|-----------|----------|
| Beauty & Fitness | Single Hill     | $-3,008 \pm 86$                   | 30.3      | 108       | 0.33     |
|                  | Mixture $K = 2$ | $-2,992 \pm 96$                   | 29.6      | 101       | 0.37     |
|                  | Mixture $K = 3$ | <b><math>-2,786 \pm 48</math></b> | 34.3      | 122       | 0.54     |
| Home & Garden    | Single Hill     | $-6,168 \pm 63$                   | 26.0      | 25.0      | 0.87     |
|                  | Mixture $K = 2$ | $-5,985 \pm 56$                   | 31.4      | 27.2      | 0.88     |
|                  | Mixture $K = 3$ | <b><math>-5,860 \pm 48</math></b> | 28.2      | 25.9      | 0.86     |
| Toys & Hobbies   | Single Hill     | $-6,379 \pm 89$                   | 36.4      | 57.3      | 0.94     |
|                  | Mixture $K = 2$ | $-6,346 \pm 89$                   | 29.3      | 53.5      | 0.98     |
|                  | Mixture $K = 3$ | <b><math>-6,104 \pm 91</math></b> | 34.3      | 47.3      | 0.96     |

Two important caveats apply to this overlay. First, the synthetic axis  $\bar{d}_{\cos}$  is computed from the *true* component curves while the real axis is computed from *posterior-mean* curves; the latter inherits the chain-mixing problems documented in Table 3 and should be read as a point estimate with substantial uncertainty for the failed fits. Second, the  $\bar{d}_{\cos} \approx 0.10$  transition in Section 6 is an empirical, model-dependent boundary rather than a sharp identifiability threshold (e.g. Mixture  $K = 3$  on the  $K_{\text{true}} = 3$  DGP recovers Shannon count  $\approx 2.27$  at true cosine 0.058).

Subject to those caveats, the picture is heterogeneous across organisations. *Home & Garden* sits at the resolvability transition: its posterior cosine separation is 0.09 under Mixture  $K = 2$  and 0.12 under Mixture  $K = 3$ , and the Shannon count  $^1D_{\text{post}}$  stays at 1.27 and 1.66 respectively. Among the three organisations this fit reaches the convergence threshold most closely (publication-ready 1/3 for  $K = 2$ , with the  $K = 3$  fit failing only at the strict  $\hat{R} \leq 1.01$  bar), so its position on the diagnostic plane is the one we are most confident reporting. *Toys & Hobbies* under Mixture  $K = 3$  has a posterior separation of 0.29 and a Shannon count of 2.72, which is inside the resolvable region. However, all three Mixture  $K = 3$  Toys & Hobbies fits exhibit large numbers of max-tree-depth saturations and tail ESS per chain of 23–34, so the posterior-mean curves driving the 0.29 figure are themselves uncertain; we therefore present this case as consistent with three components rather than as evidence for them. *Beauty & Fitness* occupies the opposite extreme: the highest posterior separation of any organisation (0.46 for Mixture  $K = 2$ , 0.68 for Mixture  $K = 3$ ), but every fit fails publication-readiness with max  $\hat{R}$  in the range 2.4–3.6 and tail ESS per chain of order 1–6 for  $K = 3$ . At that level the posterior summary itself is unreliable; the high posterior separation should be read as a flag for “the model finds widely-separated modes” rather than as a measurement of structure in the data.

The combined reading is that the diagnostic picture is organisation-dependent. Home & Garden lives at the resolvability boundary with reliable enough diagnostics that we can locate it on the plane. Toys & Hobbies is consistent with three components subject to chain-mixing caveats. Beauty & Fitness exhibits strong multi-modality in the posterior coupled with severe chain mixing failures; substantive claims about heterogeneity there should wait for a stronger inference setup or a heavier-tailed likelihood.

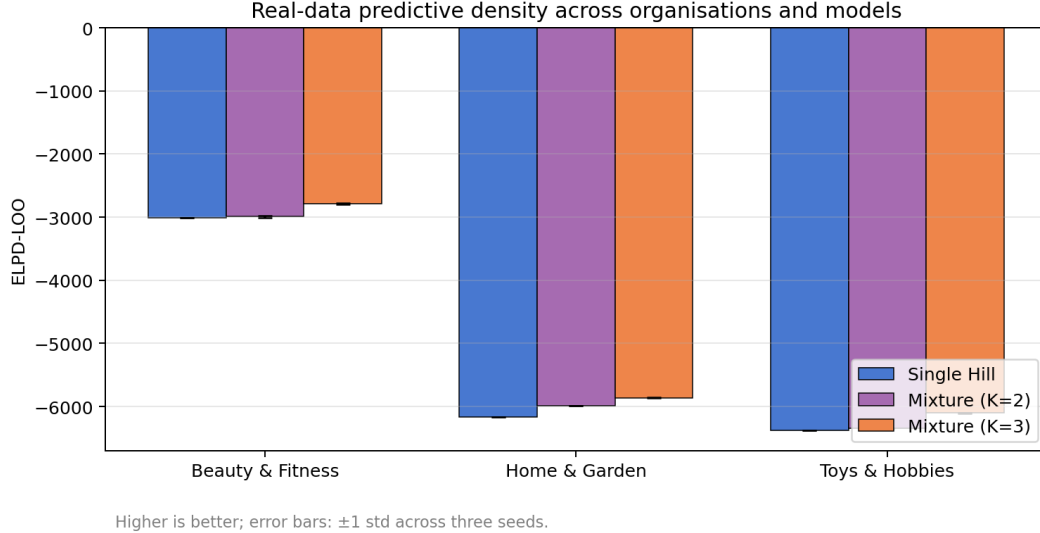


Figure 7: Real-data ELPD-LOO across the three organisations and three models. Higher is better. Error bars are  $\pm 1$  standard deviation across three NUTS reruns at different random seeds and represent Monte Carlo variability of the sampler, not data-level uncertainty; in particular they should not be read as paired `se_diff` between models. Mixture  $K = 3$  achieves the highest mean ELPD-LOO in every organisation.

## 8 Discussion

**Mixture helps, but mainly in predictive density.** Across both synthetic and real data, the most consistent benefit of mixture-of-Hill over single-Hill is in ELPD-LOO. The CRPS gains are smaller, and improvements in MAPE are organisation-specific. This pattern—predictive density improves first—is consistent with the view that mixtures primarily improve probabilistic calibration and uncertainty representation rather than mean prediction. A practitioner whose objective is point forecasting may gain little; a practitioner who needs calibrated predictive intervals for budget decisions gains more.

**The synthetic–real diagnostic gap.** The benchmark convergence threshold ( $\hat{R} \leq 1.05$ ) passes on 44/45 synthetic fits but fails on 0/9 Mixture  $K = 3$  fits on real data; the stricter publication-ready criterion ( $\hat{R} \leq 1.01$ ) passes on 33/45 synthetic fits and 0/9 real Mixture  $K = 3$  fits. This is not a contradiction: synthetic data are generated within the model class, with controlled noise and stationary mixing weights. Real organisations exhibit additional structure—seasonality, regime shifts, channel-mix changes, and outliers—that even the looser threshold is sensitive to. We report this gap rather than relaxing the criterion, because the gap itself is one of the most informative findings: it tells practitioners that synthetic-benchmark convergence does not automatically transfer.

**Effective  $K$  is organisation-dependent.** The Shannon effective counts  ${}^1D_{\text{post}}$  in Table 3 (between 1.66 and 2.72 for Mixture  $K = 3$ ) and the active component counts  $K_{\text{active}}$  (between 2.00 and 2.17) tell mutually consistent but not identical stories: Home & Garden uses two strongly-imbalanced components ( ${}^1D_{\text{post}} = 1.66$  but  $K_{\text{active}} = 2.00$ ), while Toys & Hobbies is the only organisation in which the Shannon count clears 2.5. Subject to the chain-mixing caveats reiterated above, Home & Garden sits at the resolvability boundary, Toys & Hobbies is consistent with three

Posterior response curves — Home & Garden (seed 0)

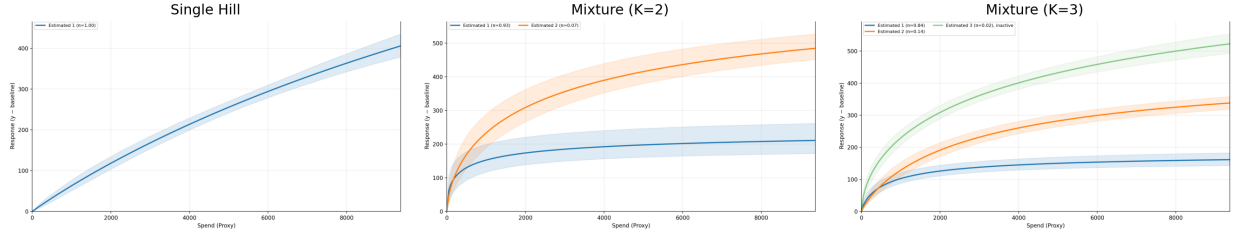


Figure 8: Posterior response curves for Home & Garden (seed 0) under Single Hill, Mixture  $K = 2$ , and Mixture  $K = 3$ . Shaded bands are 90% posterior intervals; legend entries report posterior mixture weights  $\hat{\pi}$  per component. The Mixture  $K = 3$  third component is labelled *inactive* because its weight collapses to 0.02.

components but its posterior summary itself is uncertain at the strict diagnostic level, and Beauty & Fitness shows the strongest mixture/single-Hill ELPD gap, the highest posterior separation, and the worst publication-readiness rate. The right read of these three cases is not “mixture saturates at two” but rather “mixture-of-Hill has organisation-specific identifiability behaviour, and the resolvability framework lets us name the regime each organisation falls into even when the strict criterion is not met”.

**Beauty & Fitness as a cautionary case.** The Beauty & Fitness organisation has the strongest mixture-versus-single ELPD gap but also the most striking calibration failure: 90% holdout coverage rises from 0.33 to 0.54 across models but never approaches the nominal level. The high Pareto- $k$  count further indicates influential observations. The honest reading is that the mixture is doing real work but is still not a sufficient model for this organisation, and that a heavier-tailed likelihood or richer observation model would be needed before deploying it for decisions.

**Identifiability caveat.** As Dew et al. [2024] note, nonlinear and time-varying effects in MMM are jointly under-identified from typical observational time series. Our mixture-of-Hill model does not resolve this concern; it offers a controlled extension of the standard family in which the source of nonlinearity (heterogeneity, in this case) is at least made explicit. Stronger causal claims would require external variation, such as geo-experiments or incrementality tests.

## 9 Limitations

We separate limitations from open conclusions so that each can be addressed independently in follow-up work.

**Data and scope.** The real-data evaluation covers three organisations from three verticals of one e-commerce dataset; cross-organisation generalisation claims would require a substantially larger panel. We aggregate spend across channels into a single  $x_t$  in both benchmarks, which simplifies inference but loses cross-channel information. Channel-level mixture-of-Hill is a direct extension, with the caveat that label switching now appears within *each* channel.

Table 3: Real-data convergence pass rates (publication-ready criterion across three seeds), Shannon effective count  ${}^1D_{\text{post}} = \exp(-\sum_k \hat{\pi}_k \log \hat{\pi}_k)$  evaluated at the posterior mean of  $\hat{\pi}$ , active component count  $K_{\text{active}}$  (number of mixture components with posterior mean weight above  $1/(2K)$  averaged across seeds), the mean and maximum number of LOO observations with  $\hat{k}_i > 0.7$  across seeds, and the posterior mean pairwise cosine separation between active Hill components (Section 7.4).  ${}^1D_{\text{post}}$  and  $K_{\text{active}}$  disagree because Shannon down-weights uneven mixtures: e.g. Home & Garden Mixture  $K = 3$  uses two dominant components ( $K_{\text{active}} = 2.00$ ) but with weights  $\approx 0.84, 0.14$ , so  ${}^1D_{\text{post}} = 1.66$ .

| Organisation     | Model           | Pub. pass rate | ${}^1D_{\text{post}}$ | $K_{\text{active}}$ | $\hat{k}_i > 0.7$ (mean / max) | $\bar{d}_{\text{cos}}^{\text{post}}$ |
|------------------|-----------------|----------------|-----------------------|---------------------|--------------------------------|--------------------------------------|
| Beauty & Fitness | Single Hill     | 0.67           | 1.00                  | 1.00                | 1.7 / 3                        | —                                    |
|                  | Mixture $K = 2$ | 0.00           | 1.26                  | 1.65                | 15.0 / 21                      | 0.46                                 |
|                  | Mixture $K = 3$ | 0.00           | 1.79                  | 2.17                | 3.3 / 10                       | 0.68                                 |
| Home & Garden    | Single Hill     | 0.67           | 1.00                  | 1.00                | 0.7 / 1                        | —                                    |
|                  | Mixture $K = 2$ | 0.33           | 1.27                  | 1.94                | 0.7 / 1                        | 0.09                                 |
|                  | Mixture $K = 3$ | 0.00           | 1.66                  | 2.00                | 1.7 / 3                        | 0.12                                 |
| Toys & Hobbies   | Single Hill     | 1.00           | 1.00                  | 1.00                | 2.7 / 4                        | —                                    |
|                  | Mixture $K = 2$ | 0.67           | 1.50                  | 2.00                | 1.7 / 2                        | 0.47                                 |
|                  | Mixture $K = 3$ | 0.00           | 2.72                  | 2.45                | 3.3 / 4                        | 0.29                                 |

**Model structure.** The model assumes mixing weights that are constant over time. Several of the publication-ready failures on real data are consistent with regime shifts (seasonality, channel-mix changes) that this assumption cannot absorb. The Gaussian likelihood is also not robust to influential observations, as seen most clearly in the Beauty & Fitness Pareto- $k$  counts; a Student- $t$  or heavy-tailed alternative would be a natural robustification.

**Diagnostics.** The publication-ready criterion is strict by design ( $\hat{R} \leq 1.01$ , ESS  $\geq 100$  per chain, zero divergences). Real-data fits routinely fail this bar even when posterior summaries look qualitatively reasonable. We have not yet investigated whether a softer, application-aware criterion would distinguish “failure caused by misspecification” from “failure caused by posterior multi-modality with otherwise-good summaries”.

**Priors and scales.** The prior on  $\lambda$  depends on the empirical spend range, and the prior on  $A$  on  $\text{range}(y)$ . These data-dependent priors are operationally convenient but should be made explicit in any deployment. We have not run a sensitivity sweep on the multiplicative constants in those priors.

**Causal identification.** As above, the mixture extension does not address the joint identification problem in Dew et al. [2024]. Practitioners using this model for budget decisions should triangulate against geo-experiments or incrementality tests rather than relying on observational ELPD alone.

## 10 Code and Data Availability

All code, configurations, and per-case artifacts are released at <https://github.com/jpcca/marketing-mix-model>. The repository contains: model definitions and inference utilities (`src/hill_mixture_mmm/`), the

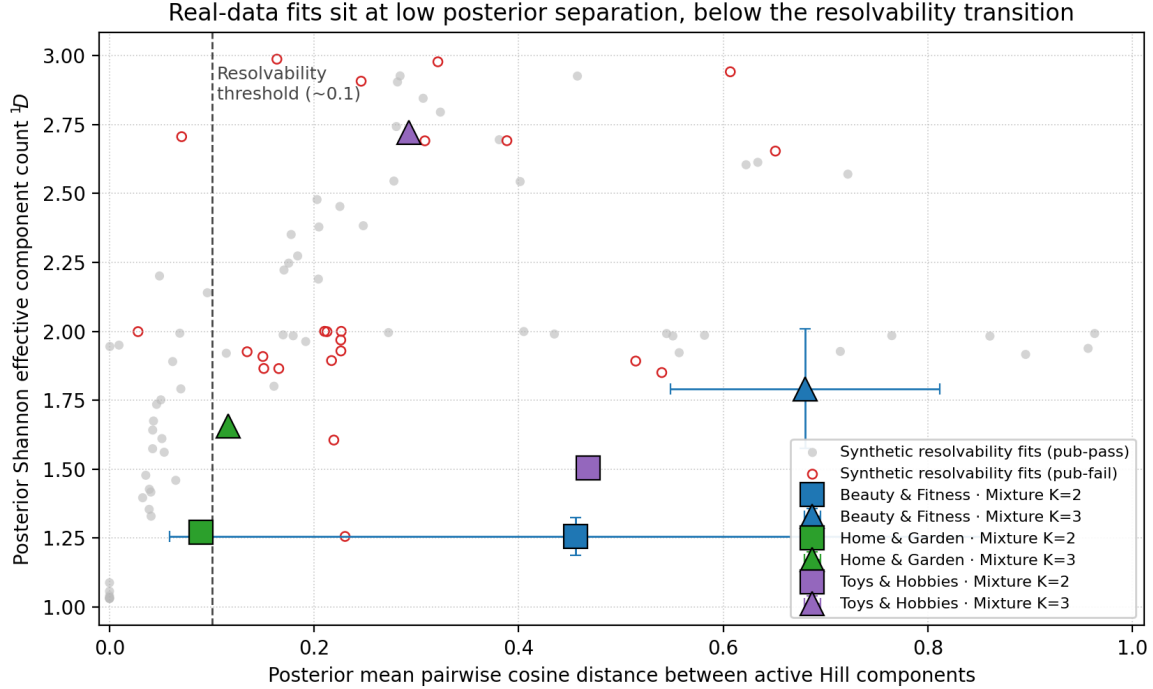


Figure 9: Posterior Shannon effective component count versus posterior mean pairwise cosine distance for the 90 synthetic resolvability fits (grey, with red-outlined fits failing the publication-ready criterion) and the three real organisations averaged across seeds (large coloured markers; error bars are  $\pm 1$  s.d. over seeds). The dashed line marks the cosine  $\approx 0.10$  resolvability transition identified in Section 6.

synthetic data generator (`data.py`), the synthetic and real-data benchmark drivers (`scripts/run_benchmark.py`, `scripts/run_real_data_validation.py`), the resolvability sweep (`scripts/run_component_resolvability_sweep.py`) and the analysis script that computes posterior cosine separation for Section 7.4 (`scripts/compute_posterior_sep.py`). Per-fit JSON summaries containing posterior weights, label-invariant and relabeled diagnostics, and convergence flags are checked in under `paper/figures/{synthetic,real}/`. Random seeds are fixed and listed alongside the per-fit artifacts.

## 11 Conclusion

We presented a Bayesian mixture of Hill saturation functions for Marketing Mix Modeling, together with a label-invariant diagnostic workflow and three complementary benchmarks. Mixture-of-Hill consistently improves ELPD-LOO under heterogeneity, with mean ELPD gains of +222 to +308 over single-Hill on the three real organisations; under a coarse paired-SE upper bound these gaps are  $\approx 2$ –4 standard errors in magnitude, and the gains appear in predictive density rather than uniformly across MAPE/CRPS—on Beauty & Fitness MAPE and CRPS in fact deteriorate slightly even as ELPD improves. The component resolvability study identifies an empirical cosine-separation transition near  $\approx 0.10$  below which the posterior tends to collapse to fewer effective components; the transition is model-dependent and should not be read as a sharp identifiability threshold. Re-using this axis on the real fits provides a diagnostic vocabulary for the three organisations: Home & Garden sits at the boundary, Toys & Hobbies is consistent with three components

subject to chain-mixing caveats, and Beauty & Fitness exhibits strong multi-modality together with severe chain mixing failures. The benchmark convergence threshold that succeeds on 44/45 synthetic fits fails uniformly for Mixture  $K = 3$  on real data; we interpret this gap as a directly useful finding for practitioners deploying mixture MMMs rather than a weakness to be sanded away.

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## A Inference Hyperparameters

Table 4 reproduces the per-model NUTS settings used throughout the paper. Two chains are run in parallel for every fit; warmup uses NUTS dual averaging with a window size matched to the warmup length. Tree depth is capped at 10, the initialisation strategy is `numpyro.infer.init_to_median`, and mass adaptation uses a diagonal mass matrix.



Table 4: Per-model NUTS configuration. Synthetic and real-data fits differ only in warmup/sampling length; all other settings are shared. The single-Hill synthetic schedule uses 600 warmup/samples on the  $K = 1$  and  $K = 2$  DGPs and 800 on the  $K = 3$  DGP; mixtures use the same schedule across DGPs.

| Model           | Warmup (synthetic / real) | Samples (synthetic / real) | Chains | Target accept |
|-----------------|---------------------------|----------------------------|--------|---------------|
| Single Hill     | 600–800 / 600             | 600–800 / 600              | 2      | 0.90          |
| Mixture $K = 2$ | 900 / 900                 | 900 / 900                  | 2      | 0.95          |
| Mixture $K = 3$ | 1,600 / 1,600             | 1,200 / 1,200              | 2      | 0.97          |

## B Priors

Table 5 lists the priors used in every experiment. The priors on  $A_k$  and  $\mu_0$  scale automatically with the training-data range and mean to accommodate diverse target magnitudes; the prior on  $\lambda$  uses a cumulative-sum reparameterization (Section 3.1) with positive log-normal increments. We have not run a formal sensitivity sweep on the multiplicative constants; that is noted as a limitation in Section 9.

Table 5: Priors. Statistics with hats (e.g.  $\widehat{\text{range}}(y)$ ,  $\bar{y}$ ,  $\hat{\sigma}_y$ ) are computed from the training portion only.

| Parameter                             | Distribution                                                  | Notes                                              |
|---------------------------------------|---------------------------------------------------------------|----------------------------------------------------|
| Adstock decay $\alpha$                | Beta(2, 2)                                                    | shared across components                           |
| Half-saturation increments $\delta_j$ | LogNormal( $\log(s_{\max}/(K+1))$ , 0.7)                      | $\lambda_k = \lambda_1 + \sum_{j \leq k} \delta_j$ |
| Hill slope $n_k$                      | LogNormal( $\log 1.5$ , 0.4)                                  | per component                                      |
| Component amplitude $A_k$             | LogNormal( $\log(0.3 \cdot \widehat{\text{range}}(y))$ , 0.8) | per component                                      |
| Mixture weights $\pi$                 | Dirichlet( $\mathbf{1}_K$ )                                   | symmetric                                          |
| Intercept $\mu_0$                     | $\mathcal{N}(\bar{y}, 2\hat{\sigma}_y)$                       | data-dependent                                     |
| Trend $\beta$                         | $\mathcal{N}(0, 0.5)$                                         | on standardised $t$                                |
| Observation noise $\sigma$            | HalfNormal( $\hat{\sigma}_y$ )                                | data-dependent                                     |

## C Synthetic DGP Parameters

Table 6 summarises the parameter ranges used to generate synthetic data for the three DGPs in Section 5 and the nine profiles used in Section 6. Per-component values are drawn uniformly from the listed ranges and held fixed across the five seeds within each profile.

Table 6: Synthetic DGP parameter ranges. The resolvability profiles fix  $A_k$  and  $n_k$  and vary only the spacing  $\delta$  of the half-saturation points  $\lambda_k$  around a fixed centre  $c$ .

| Parameter   | Single ( $K = 1$ ) | Mixture $K = 2$ | Mixture $K = 3$ |
|-------------|--------------------|-----------------|-----------------|
| $\alpha$    | 0.5                | 0.5             | 0.5             |
| $\lambda_k$ | [1.0, 2.5]         | [0.8, 2.8]      | [0.7, 3.0]      |
| $n_k$       | [1.0, 2.0]         | [1.0, 2.0]      | [1.0, 2.0]      |
| $A_k$       | [80, 120]          | [80, 130]       | [80, 140]       |
| $\pi_k$     | 1                  | {0.6, 0.4}      | {0.5, 0.3, 0.2} |
| $\sigma$    | 5.0                | 5.0             | 5.0             |
| $T$         | 200                | 200             | 200             |
| Train ratio | 0.75               | 0.75            | 0.75            |